

Week -2

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Question :

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

Task to be Done

1. **The Catalog Search (JOINS):** Create a third table called Loans that connects member_id to book_id. Write a query to show the Member Name and the Book Title they have currently borrowed.
2. **Collection Stats (Aggregations):** Write a query to find the total number of books published in each year. o Hint: Use COUNT(book_id) and GROUP BY published_year.
3. **The Donation Transaction (ACID):** Use a Transaction (BEGIN...COMMIT) to add a new book to the Books table and simultaneously log the entry in a Donation_History table.
4. **Search Speed (Indexing):** Create an Index on the isbn column. Explain why searching by ISBN is now faster for the librarian than searching by Title. Skill Set Gained • Relational Logic: Connecting independent tables (Books + Members) through a Junction table (Loans). • Analytical SQL: Using GROUP BY to generate business insights. • Atomic Operations: Using Transactions to ensure multiple database changes succeed or fail as a single unit.

Step 1 — Create Database

Limit to 1000 rows

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
1 -- Step 1: Create Database
2 CREATE DATABASE Donation_db;
3 USE Donation_db;
4
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	14:34:32	CREATE DATABASE Donation_db	1 row(s) affected	0.016 sec
✓ 2	14:34:38	USE Donation_db	0 row(s) affected	0.000 sec

Step 2 — Create Books Table (DDL + Constraints)

Limit to 1000 rows

```
1 -- Step 1: Create Database
2 CREATE DATABASE Donation_db;
3 USE Donation_db;
4 -- Step 2: Create Books Table
5 CREATE TABLE Books(
6     book_id INT PRIMARY KEY,
7     title VARCHAR(100) NOT NULL,
8     isbn VARCHAR(20) UNIQUE,
9     published_year INT CHECK(published_year < 2027)
10 );
11
```

Output

Action Output

#	Time	Action	Message
✓ 1	14:34:32	CREATE DATABASE Donation_db	1 row(s) affected
✓ 2	14:34:38	USE Donation_db	0 row(s) affected
✓ 3	18:13:07	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) ...	0 row(s) affected

```
1 -- Step 1: Create Database
2 • CREATE DATABASE Donation_db;
3 • USE Donation_db;
4 -- Step 2: Create Books Table
5 • CREATE TABLE Books(
6     book_id INT PRIMARY KEY,
7     title VARCHAR(100) NOT NULL,
8     isbn VARCHAR(20) UNIQUE,
9     published_year INT CHECK(published_year < 2027)
10 );
11 -- Step 3: Insert Records into Books Table
12 • INSERT INTO Books VALUES
13 (1,'The Great Gatsby','9780743273565',1925),
14 (2,'To Kill a Mockingbird','9780061120084',1960),
15 (3,'1984','9780451524935',1949);
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	14:34:32	CREATE DATABASE Donation_db	1 row(s) affected
✓ 2	14:34:38	USE Donation_db	0 row(s) affected
✓ 3	18:13:07	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) ...	0 row(s) affected
✓ 4	18:14:23	INSERT INTO Books VALUES (1,'The Great Gatsby','9780743273565',1925), (2,'To Kill a Mockingbird','9780061120...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

```
7     title VARCHAR(100) NOT NULL,
8     isbn VARCHAR(20) UNIQUE,
9     published_year INT CHECK(published_year < 2027)
10 );
11 -- Step 3: Insert Records into Books Table
12 • INSERT INTO Books VALUES
13 (1,'The Great Gatsby','9780743273565',1925),
14 (2,'To Kill a Mockingbird','9780061120084',1960),
15 (3,'1984','9780451524935',1949);
16 -- Step 4: Display Books Table
17 • SELECT * FROM Books;
```

Result Grid				
Filter Rows:				
	book_id	title	isbn	published_year
▶	1	The Great Gatsby	9780743273565	1925
	2	To Kill a Mockingbird	9780061120084	1960
	3	1984	9780451524935	1949
•	NULL	NULL	NULL	NULL

Books 1 x			
Output			
Action Output			
#	Time	Action	Message
✓ 1	14:34:32	CREATE DATABASE Donation_db	1 row(s) affected
✓ 2	14:34:38	USE Donation_db	0 row(s) affected
✓ 3	18:13:07	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(...	0 row(s) affected
✓ 4	18:14:23	INSERT INTO Books VALUES (1,'The Great Gatsby','9780743273565',1925), (2,'To Kill a Mockingbird','978006...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 5	18:16:31	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned

```

12 • INSERT INTO Books VALUES
13 (1,'The Great Gatsby','9780743273565',1925),
14 (2,'To Kill a Mockingbird','9780061120084',1960),
15 (3,'1984','9780451524935',1949);
16 -- Step 4: Display Books Table
17 • SELECT * FROM Books;
18 -- Step 5: Create Members Table
19 • CREATE TABLE Members(
20     member_id INT PRIMARY KEY,
21     full_name VARCHAR(100),
22     email VARCHAR(100) UNIQUE
23 );
24

```

Output

Action Output

#	Time	Action	Message
✓	1 14:34:32	CREATE DATABASE Donation_db	1 row(s) affected
✓	2 14:34:38	USE Donation_db	0 row(s) affected
✓	3 18:13:07	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(100) NOT NULL, year INT NOT NULL);	0 row(s) affected
✓	4 18:14:23	INSERT INTO Books VALUES (1,'The Great Gatsby','9780743273565',1925), (2,'To Kill a Mockingbird','9780061120084',1960), (3,'1984','9780451524935',1949);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	5 18:16:31	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓	6 18:17:49	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	0 row(s) affected

```

23 );
24 -- Step 6: Insert Records into Members Table
25 • INSERT INTO Members VALUES
26 (101,'Harshika','iliharshika@gmail.com'),
27 (102,'Bhanu Priya','bhanupriya@gmail.com'),
28 (103,'Nikshitha','nikshitha@gmail.com');

```

Output

Action Output

#	Time	Action	Message
✓	5 18:16:31	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓	6 18:17:49	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	0 row(s) affected
✗	7 18:19:17	-- Step 6: Insert Records into Members Table INSERT INTO Members VALUES (101,'Harshika','nikshitha@gmail.com'), (102,'Bhanu Priya','bhanupriya@gmail.com'), (103,'Nikshitha','nikshitha@gmail.com');	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'nikshitha@gmail.com' at line 1
✗	8 18:19:49	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	Error Code: 1050. Table 'members' already exists
✗	9 18:20:20	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(100) NOT NULL, year INT NOT NULL);	Error Code: 1050. Table 'books' already exists
✓	10 18:20:59	INSERT INTO Members VALUES (101,'Harshika','iliharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gmail.com'), (103,'Nikshitha','nikshitha@gmail.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

```

29 -- Step 7: Display Members Table
30 • SELECT * FROM Members;

```

Result Grid

member_id	full_name	email
101	Harshika	iliharshika@gmail.com
102	Bhanu Priya	bhanupriya@gmail.com
103	Nikshitha	nikshitha@gmail.com

Members 2 x

Output

Action Output

#	Time	Action	Message
✓	6 18:17:49	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	0 row(s) affected
✗	7 18:19:17	-- Step 6: Insert Records into Members Table INSERT INTO Members VALUES (101,'Harshika','nikshitha@gmail.com'), (102,'Bhanu Priya','bhanupriya@gmail.com'), (103,'Nikshitha','nikshitha@gmail.com');	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'nikshitha@gmail.com' at line 1
✗	8 18:19:49	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);	Error Code: 1050. Table 'members' already exists
✗	9 18:20:20	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(100) NOT NULL, year INT NOT NULL);	Error Code: 1050. Table 'books' already exists
✓	10 18:20:59	INSERT INTO Members VALUES (101,'Harshika','iliharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gmail.com'), (103,'Nikshitha','nikshitha@gmail.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	11 18:21:54	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned

```

29 -- Step 7: Display Members Table
30 • SELECT * FROM Members;
31 -- Step 8: Create Loans Table
32 • CREATE TABLE Loans(
33     loan_id INT PRIMARY KEY,
34     member_id INT,
35     book_id INT,
36     loan_date DATE,
37     FOREIGN KEY(member_id) REFERENCES Members(member_id),
38     FOREIGN KEY(book_id) REFERENCES Books(book_id)
39 );
40

```

Context Help

Output				
Action Output				
#	Time	Action	Message	
7	18:19:17	- Step 6: Insert Records into Members Table INSERT INTO Members VALUES (101,'Harshika','nikshtha@gma...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	
8	18:19:49	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCH...	Error Code: 1050. Table 'members' already exists	
9	18:20:20	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHA...	Error Code: 1050. Table 'books' already exists	
10	18:20:59	INSERT INTO Members VALUES (101,'Harshika','lharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	
11	18:21:54	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	
12	18:22:50	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected	

```

39 );
40 -- Step 9: Insert Records into Loans Table
41 • INSERT INTO Loans VALUES
42 (1,101,1,'2025-01-05'),
43 (2,102,2,'2025-01-08'),
44 (3,103,3,'2025-01-10'),
45 (4,101,2,'2025-02-01'),
46 (5,102,1,'2025-02-05'),
47 (6,103,2,'2025-02-12'),
48 (7,101,3,'2025-03-01'),
49 (8,102,3,'2025-03-07'),
50 (9,103,1,'2025-03-15'),
51 (10,101,1,'2025-04-01');
52

```

Output				
Action Output				
#	Time	Action	Message	
8	18:19:49	CREATE TABLE Members(member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCH...	Error Code: 1050. Table 'members' already exists	
9	18:20:20	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHA...	Error Code: 1050. Table 'books' already exists	
10	18:20:59	INSERT INTO Members VALUES (101,'Harshika','lharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	
11	18:21:54	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	
12	18:22:50	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected	
13	18:24:48	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'2025-01-10'), (4,101,2,'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	

```

46 (5,102,1,'2025-02-05'),
47 (6,103,2,'2025-02-12'),
48 (7,101,3,'2025-03-01'),
49 (8,102,3,'2025-03-07'),
50 (9,103,1,'2025-03-15'),
51 (10,101,1,'2025-04-01');
52 -- Step 10: Display Loans Table
53 • SELECT * FROM Loans;

```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	loan_id	member_id	book_id	loan_date
▶	1	101	1	2025-01-05
	2	102	2	2025-01-08
	3	103	3	2025-01-10
	4	101	2	2025-02-01
	5	102	1	2025-02-05
	6	103	2	2025-02-12
	7	101	3	2025-03-01
	8	102	3	2025-03-07
	9	103	1	2025-03-15
	10	101	1	2025-04-01
•	NULL	NULL	NULL	NULL

Loans 3 x

Output

Action Output

#	Time	Action	Message
9	18:20:20	CREATE TABLE Books(book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	Error Code: 1050. Table 'books' already exists
10	18:20:59	INSERT INTO Members VALUES (101,'Harshika','liharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
11	18:21:54	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
12	18:22:50	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected
13	18:24:48	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'2025-01-10'), (4,101,2,'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
14	18:25:44	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned

```

52 -- Step 10: Display Loans Table
53 • SELECT * FROM Loans;
54 -- Step 11: Display Member Name and Borrowed Book Using INNER JOIN
55 • SELECT
56     m.full_name AS Member_Name,
57     b.title AS Book_Title
58 FROM Loans l
59 INNER JOIN Members m
60 ON l.member_id = m.member_id
61 INNER JOIN Books b
62 ON l.book_id = b.book_id;

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	Member_Name	Book_Title
▶	Harshika	The Great Gatsby
	Harshika	To Kill a Mockingbird
	Harshika	1984
	Harshika	The Great Gatsby
	Bhanu Priya	To Kill a Mockingbird
	Bhanu Priya	The Great Gatsby
	Bhanu Priya	1984
	Nikshitha	1984
	Nikshitha	To Kill a Mockingbird
	Nikshitha	The Great Gatsby

Result 4 x

Output

Action Output

#	Time	Action	Message
10	18:20:59	INSERT INTO Members VALUES (101,'Harshika','liharshika@gmail.com'), (102,'Bhanu Priya','bhanupriya@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
11	18:21:54	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
12	18:22:50	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected
13	18:24:48	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'2025-01-10'), (4,101,2,'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
14	18:25:44	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
15	18:26:57	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m O...	10 row(s) returned


```

60 ON l.member_id = m.member_id
61 INNER JOIN Books b
62 ON l.book_id = b.book_id;
63 -- Step 12: Count Books Published Each Year Using GROUP BY
64 • SELECT
65     published_year,
66     COUNT(book_id) AS Total_Books
67 FROM Books
68 GROUP BY published_year;

```

published_year	Total_Books
1925	1
1960	1
1949	1

Result 5 x

Output

Action Output

#	Time	Action	Message
✓ 11	18:21:54	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
✓ 12	18:22:50	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected
✓ 13	18:24:48	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'2025-01-10'), (4,101,2,'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 14	18:25:44	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
✓ 15	18:26:57	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m O...	10 row(s) returned
✓ 16	18:27:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year LI...	3 row(s) returned

```

63 -- Step 12: Count Books Published Each Year Using GROUP BY
64 • SELECT
65     published_year,
66     COUNT(book_id) AS Total_Books
67 FROM Books
68 GROUP BY published_year;
69 -- Step 13: Create Donation History Table
70 • CREATE TABLE Donation_History(
71     donation_id INT PRIMARY KEY,
72     book_id INT,
73     donor_name VARCHAR(100),
74     donation_date DATE,
75     FOREIGN KEY(book_id) REFERENCES Books(book_id)
76 );
77

```

Output

Action Output

#	Time	Action	Message
✓ 12	18:22:50	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected
✓ 13	18:24:48	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'2025-01-10'), (4,101,2,'...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 14	18:25:44	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
✓ 15	18:26:57	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m O...	10 row(s) returned
✓ 16	18:27:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year LI...	3 row(s) returned
✓ 17	18:28:39	CREATE TABLE Donation_History(donation_id INT PRIMARY KEY, book_id INT, donor_name VARC...	0 row(s) affected

```

69 -- Step 13: Create Donation History Table
70 CREATE TABLE Donation_History(
71     donation_id INT PRIMARY KEY,
72     book_id INT,
73     donor_name VARCHAR(100),
74     donation_date DATE,
75     FOREIGN KEY(book_id) REFERENCES Books(book_id)
76 );
77 -- Step 14: Start Transaction
78 START TRANSACTION;

```

Output

Action Output

#	Time	Action	Message
✓ 13	18:24:48	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'2025-01-08'), (3,103,3,'2025-01-10'), (4,101,2,'...)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 14	18:25:44	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
✓ 15	18:26:57	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id	10 row(s) returned
✓ 16	18:27:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year LIMIT 0, 3	3 row(s) returned
✓ 17	18:28:39	CREATE TABLE Donation_History(donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date DATE, FOREIGN KEY(book_id) REFERENCES Books(book_id))	0 row(s) affected
✓ 18	18:29:25	START TRANSACTION	0 row(s) affected

```

76 );
77 -- Step 14: Start Transaction
78 START TRANSACTION;
79 -- Step 15: Insert New Book into Books Table
80 INSERT INTO Books VALUES
81 (4,'Animal Farm','9780451526342',1945);

```

Output

Action Output

#	Time	Action	Message
✓ 14	18:25:44	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
✓ 15	18:26:57	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id	10 row(s) returned
✓ 16	18:27:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year LIMIT 0, 3	3 row(s) returned
✓ 17	18:28:39	CREATE TABLE Donation_History(donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date DATE, FOREIGN KEY(book_id) REFERENCES Books(book_id))	0 row(s) affected
✓ 18	18:29:25	START TRANSACTION	0 row(s) affected
✓ 19	18:30:56	START TRANSACTION	0 row(s) affected

```

78 START TRANSACTION;
79 -- Step 15: Insert New Book into Books Table
80 INSERT INTO Books VALUES
81 (4,'Animal Farm','9780451526342',1945);
82 -- Step 16: Insert Donation Record and Commit Transaction
83 INSERT INTO Donation_History VALUES
84 (1,4,'Raj Kumar',CURDATE());
85

```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 15	18:26:57	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id	10 row(s) returned
✓ 16	18:27:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year LIMIT 0, 3	3 row(s) returned
✓ 17	18:28:39	CREATE TABLE Donation_History(donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date DATE, FOREIGN KEY(book_id) REFERENCES Books(book_id))	0 row(s) affected
✓ 18	18:29:25	START TRANSACTION	0 row(s) affected
✓ 19	18:30:56	START TRANSACTION	0 row(s) affected
✗ 20	18:31:59	INSERT INTO Donation_History VALUES (1,4,'Raj Kumar',CURDATE())	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('donation_db`.`donation_hist...


```

82 -- Step 16: Insert Donation Record and Commit Transaction
83 • INSERT INTO Donation_History VALUES
84 (1,4,'Raj Kumar',CURDATE());
85 • COMMIT;
86 -- Step 17: Create Index on ISBN Column
87 • CREATE INDEX idx_isbn
88 ON Books(isbn);

```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 16	18:27:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year LI...	3 row(s) returned
✓ 17	18:28:39	CREATE TABLE Donation_History(donation_id INT PRIMARY KEY, book_id INT, donor_name VARC...	0 row(s) affected
✓ 18	18:29:25	START TRANSACTION	0 row(s) affected
✓ 19	18:30:56	START TRANSACTION	0 row(s) affected
✗ 20	18:31:59	INSERT INTO Donation_History VALUES (1,4,'Raj Kumar',CURDATE())	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('donation_db','donation_hist...
✓ 21	18:32:52	CREATE INDEX idx_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

```

86 -- Step 17: Create Index on ISBN Column
87 • CREATE INDEX idx_isbn
88 ON Books(isbn);
89 -- Step 18: Search Book Using ISBN
90 • SELECT *
91 FROM Books
92 WHERE isbn='9780451524935';

```

toggle

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

book_id	title	isbn	published_year
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

Books 6 x

Apply

Revert

Context Help

Output

Action Output

#	Time	Action	Message
✓ 17	18:28:39	CREATE TABLE Donation_History(donation_id INT PRIMARY KEY, book_id INT, donor_name VARC...	0 row(s) affected
✓ 18	18:29:25	START TRANSACTION	0 row(s) affected
✓ 19	18:30:56	START TRANSACTION	0 row(s) affected
✗ 20	18:31:59	INSERT INTO Donation_History VALUES (1,4,'Raj Kumar',CURDATE())	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('donation_db','donation_hist...
✓ 21	18:32:52	CREATE INDEX idx_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 22	18:33:29	SELECT * FROM Books WHERE isbn='9780451524935' LIMIT 0, 1000	1 row(s) returned